

StepForge Frontend PRD (Ultra Detailed - TestSprite Ready)

1. Overview

StepForge is a premium Android fitness application built with Kotlin and Jetpack Compose.

The frontend must deliver a visually stunning, high-performance UI with dynamic data-driven components.

2. Core Objectives

- Provide real-time step tracking UI
- Deliver premium visual experience
- Enable AI-driven insights visualization
- Ensure smooth performance and zero crashes

3. Screens

3.1 Home Screen

- Hero Step Card
- Animated circular progress
- Current steps
- Goal steps
- Percentage progress
- Micro animations on update
- Quick Stats Row
- Distance (meters/km)
- Calories burned
- Active time
- AI Insight Preview
- Single-line dynamic insight
- Clickable → opens detail modal

3.2 Weekly Insight Card

Collapsed:

- Weekly progress %

- Mini bar chart (7 days)
- Short AI insight

Expanded (Scrollable Modal):

- Weekly Summary (total, avg, best, worst)
- Daily Breakdown (7 days detailed)
- Performance Trend Graph
- AI Insights (3–5 generated insights)
- Recommendations (behavior-based tips)

4. UI System

- Dark theme (#0B0B0E)
- Neon cyan & green highlights
- Glassmorphism cards
- Rounded corners (16–24dp)
- Smooth animations (fade, scale)

5. Data Handling

- Use state flows
- Cache latest values
- Avoid blocking calls on UI thread
- Sync with backend in UTC

6. Edge Cases

- Zero steps → special UI
- Missing data → skeleton loader
- Goal = 0 → fallback to default

7. Performance

- Lazy loading for lists
- Minimize recomposition
- Avoid heavy operations on main thread

8. Testing Requirements (for TestSprite)

- Validate UI rendering with different step values
- Test edge cases (0, very high steps)
- Validate date consistency (UTC vs local)
- Ensure no crashes on lifecycle changes
- Validate scrollable modal performance
- Check notification sync with UI

9. Monetization UI

- Lock premium sections
- Show blurred preview
- Trigger upgrade CTA

10. Animation Requirements

- Smooth transitions
- Gesture-based expansion
- Micro feedback on user actions

End of PRD.